

Assignment 5
30 points

Stochastic Calculus
for Finance

Due Sunday, April 26

Here is the list of problems from Shreve Part II that were discussed in class (April 21, 2 pm -5 pm). The solutions to these problems were discussed in class; these have to be written up and submitted. The solutions must include the important points discussed during the class.

The assignment has to be handwritten (written on paper or an iPad or another writing tablet).

1. Exercise 4.20 (Local time)
2. Exercise 4.21 (Stop-loss start-gain paradox)
3. Exercise 6.3 (Solution of the Hull-White model)
4. Exercise 6.4 (Solution of Cox-Ingersoll-Ross model)
5. Exercise 6.6 - parts i, ii, iii (Moment-generating function for Cox-Ingersoll-Ross process)